1. Game concept and gameplay

Objective & Rules: This project is a 2.5D top-down survival arena shooter. The player controls a humanoid character and must continuously move within a closed flat arena to dodge and defeat enemies spawning from all directions. The camera is fixed relative to the player and follows their movement. There will be two to three types of monsters featuring different textures and shapes. Defeated enemies drop experience points (EXP), which are represented by small glowing green spheres that are magnetically attracted to the player when they get close. The player collects EXP to level up, which prompts a choice to receive a random weapon upgrade.

Controls & Interaction: Use WASD keys for 8-directional character movement and the mouse to aim and shoot. The mouse crosshair is decoupled from the character, utilizing a direct click-to-shoot mechanism. The core interaction loop is "move/dodge - shoot/kill - collect EXP/weapons - level up/upgrade".

Game State & Configurations: Victory is achieved by surviving for a specified time, indicated by a UI countdown timer; defeat occurs when the player's HP reaches zero.

Challenge Tasks:

1. BRDF Material Representation (map design incorporating complex materials like sand and water). 2. Normal Mapping for the ground. 3. Particle Effects for weapon hits.

2. Visual and technical design

Voxel Scene & Camera: Construct a voxel world with dimensions of at least 32×32×32 cells. The game features two level configurations: a Day level and a Night level, differentiated by their lighting setups and obstacle distributions. Utilize an isometric/top-down camera fixed above the character to provide a global view of the arena.

Lighting & Materials: Implement a physical lighting model in GLSL, including ambient, diffuse, and specular lighting. Use different specular and shininess parameters to distinguish the material appearances of metallic weapons versus wooden/flesh enemies.

Hierarchy & Animation: The player character and main monsters will use hierarchical models. Implement relative animations such as limb swinging during movement and weapon recoil during shooting.

3. Responsibilities

Member A: Responsible for core gameplay and game state management. Implement 8-directional player movement and click-to-shoot logic, enemy spawn system and chase AI, collision detection, EXP collection (magnetic attraction), level-up system (random weapon selection), and win/loss conditions (UI countdown timer). **Challenge Task (Particle Effects):** Integrate with the weapon system to create particle burst effects upon weapon hits. Weapons will be differentiated entirely by visual particle effects rather than having separate 3D models.

Member B: Responsible for world generation and base rendering. Construct the map layouts, implement UV mapping, ambient, and basic lighting. Handle the lighting configurations for the two levels: a directional light for the Day level, and a point light attached to the player for the Night level (creating a vignette effect with limited visibility). Implement at least two different material appearances and basic rendering optimization (e.g., face culling). **Challenge Task (Normal Mapping):** Enhance the geometric bump details of the flat arena floor and obstacles.

Member C: Responsible for character animation and system integration. Build the hierarchical structure and animations for multi-part characters, set up the top-down camera, HUD, and UI interfaces, and conduct the 240-frame performance benchmark. Implement instanced rendering as a core optimization to improve drawing performance and extract comparison data for the benchmark. **Challenge Task (BRDF Materials):** Implement BRDF complex materials to accurately render advanced terrain features like sand and water surfaces.

4. Three milestones

Milestone 1 (By Week 7): Complete basic voxel map rendering, camera setup, player WASD movement, one type of enemy spawning/chasing, and basic bullet shooting/collision detection.

Milestone 2 (By Week 8): Finish the EXP/level-up system, UI menus, hierarchical models and animations, complete materials and lighting (Day/Night setups), and implement the two different level configuration layouts.

Milestone 3 (By Week 9): Integrate the three challenge tasks (Particles, Normal Mapping, BRDF); complete the benchmark and record baseline vs. optimized comparison data; clean up the code and submit the final report.